



Movie Rental

Data Warehouse Design

Course	Data Warehouse
Topic	From OLTP to Dimensional Model and ETL Design
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1. Introduction

This assignment designs a high-level data warehouse for a movie rental business. The OLTP source for this assignment is the database provided in `fileDB.sql`, which is a MySQL dump of the Sakila sample database. After importing it as `sakila_prof`, the source contains 16 tables and the full operational dataset (599 customers, 1000 films, 4581 inventory items, 16,043 rentals, and 16,043 payments). This OLTP database supports daily activities such as customer registration, film rentals, payments, inventory, staff, and store information.

An OLTP system is good for fast inserts and updates, but it is not good for reporting and analysis. Business managers need a data warehouse to study rental activity, revenue, customer behavior, film popularity, store performance, and trends over time.

A data warehouse uses dimensional modeling. Data is reorganized into fact tables that store business events and measures, and dimension tables that store descriptive information. This makes analytical queries easier and faster.

1.1 OLTP vs Data Warehouse

Aspect	OLTP (Sakila)	Data Warehouse (movie_dw)
Purpose	Daily transactions	Reporting and analysis
Design	Normalized tables	Star schema (facts and dimensions)
Best for	Insert, update, delete	Aggregations and GROUP BY queries
Data	Current state	Historical and time-based
Users	Staff, applications	Managers, analysts

2. Business Questions

The data warehouse is designed to answer the following analytical business questions. These questions guided the choice of fact tables, dimensions, and measures.

#	Business Question	Why it is Important
1	Which films are rented most frequently?	Shows which films are most popular with customers.
2	Which films generate the highest revenue?	Shows which films bring in the most money.
3	Which customers rent the most films?	Identifies loyal and active customers.
4	Which customers generate the highest revenue?	Useful for customer segmentation and loyalty programs.
5	How does rental activity change over time?	Shows daily, monthly, and seasonal rental trends.
6	How does revenue change by month, quarter, or year?	Supports financial planning and forecasting.
7	How is inventory distributed between stores?	Helps manage stock levels.

3. Dimensional Model Design

3.1 Selected Business Processes

Three main business processes were chosen from the OLTP system. Each one becomes a fact table in the data warehouse:

- Rental transactions — every time a customer rents a film.
- Payment transactions — every time a customer pays for a rental.
- Inventory — the film copies that each store has.

A star schema design was chosen. In a star schema, each fact table is in the center and connects directly to dimension tables. This is simpler than a snowflake schema and easier for users and reporting tools to work with. Geographic information (city and country) is stored inside DimCustomer and DimStore instead of being separate dimensions, which is normal for a star schema.

3.2 Fact Tables

FactRental

Business Process	Film rental transactions
Grain	One row per rental transaction
Measures	Rental_Count, Rental_Duration, Late_Return_Flag
Dimensions	DimCustomer, DimFilm, DimStore, DimStaff, DimDate (rental date and return date)
Purpose	Used to analyze rental activity, film popularity, customer behavior, and late returns.

FactPayment

Business Process	Payment transactions
Grain	One row per payment
Measures	Payment_Amount
Dimensions	DimCustomer, DimFilm, DimStore, DimStaff, DimDate (payment date)
Purpose	Used to analyze revenue by film, customer, store, staff, and time.

FactInventory

Business Process	Inventory management
Grain	One row per inventory item (film copy at a store)
Measures	Inventory_Count
Dimensions	DimFilm, DimStore
Purpose	Used to analyze inventory levels and how stock is distributed between stores.

3.3 Dimension Tables

DimCustomer

Attributes	Customer_Key (PK), Customer_ID, Full_Name, Email, City, Country, Active_Status
Source OLTP Tables	customer, address, city, country
Purpose	Describes customers and supports analysis of customer behavior and geographic activity.

DimFilm

Attributes	Film_Key (PK), Film_ID, Film_Title, Category, Language, Rental_Rate, Rental_Duration, Film_Length, Release_Year, Inventory_Status
Source OLTP Tables	film, film_category, category, language, inventory
Purpose	Describes films and supports category, language, and pricing analysis. Rental_Duration is the expected duration, used to detect late returns.

DimStore

Attributes	Store_Key (PK), Store_ID, Store_Manager, City, Country
Source OLTP Tables	store, staff, address, city, country
Purpose	Describes stores and supports store performance analysis by location.

DimStaff

Attributes	Staff_Key (PK), Staff_ID, Staff_Name, Store_Assignment, Active_Status
Source OLTP Tables	staff, store
Purpose	Describes employees and supports staff performance analysis.

DimDate

Attributes	Date_Key (PK), Full_Date, Day, Month, Month_Name, Quarter, Year, Day_Of_Week, Is_Weekend
Source OLTP Tables	Generated during ETL from rental_date, return_date, and payment_date
Purpose	Shared time dimension used by all fact tables. Supports time-based and seasonal analysis.

3.4 Shared Dimensions

Some dimensions are shared between fact tables. DimCustomer, DimFilm, DimStore, and DimStaff are used by both FactRental and FactPayment. DimDate is used by all three fact tables. DimFilm and DimStore are also used by FactInventory. Sharing dimensions makes reports consistent across different fact tables.

4. Dimensional Model Diagram

The diagram below shows the complete star schema. The three fact tables are in the center and connect to the dimension tables. Each fact table has foreign keys (FK) that reference the primary keys (PK) of the dimensions. The relationships are one-to-many: each row in a dimension can be referenced by many rows in a fact table.

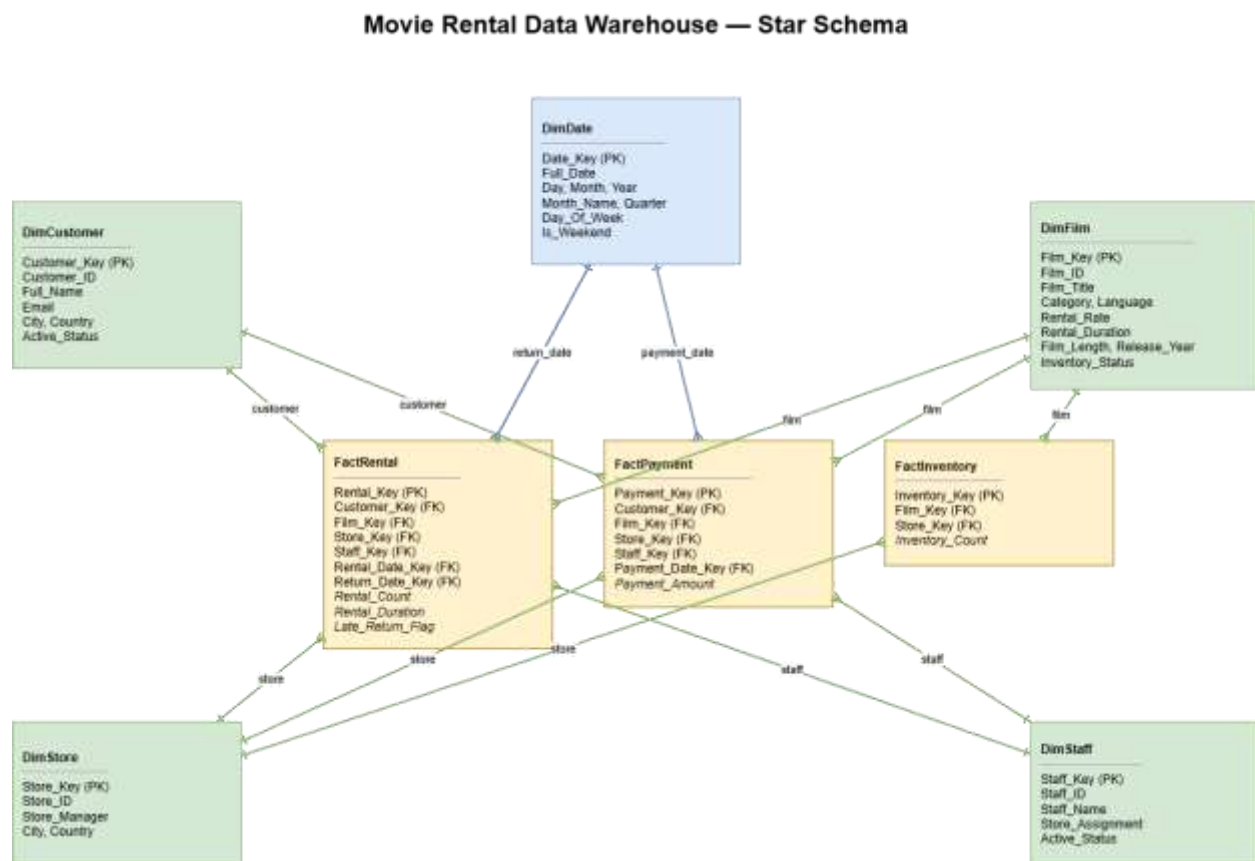


Figure 1: Movie Rental Data Warehouse Star Schema

FactRental references DimDate two times: once for the rental date and once for the return date. The Return_Date_Key can be null because some films may not be returned yet.

5. ETL Design

The ETL process moves data from the Sakila OLTP database into the movie_dw data warehouse. ETL stands for Extract, Transform, and Load. The process is implemented in Python using pandas and SQLAlchemy.

5.1 Extract

The following OLTP tables are extracted from Sakila. Each table is needed either to build a fact table or to provide descriptive information for a dimension.

Source Table	Why it is Needed
rental	Used to build FactRental. Contains rental events with dates.
payment	Used to build FactPayment. Contains payment amounts and dates.
inventory	Used to build FactInventory. Also used to find which film and store a rental belongs to.
customer	Used to build DimCustomer.
film	Used to build DimFilm. Provides title, rental rate, length, and expected duration.
film_category, category	Provide category information for DimFilm.
language	Provides language information for DimFilm.
store	Used to build DimStore.
staff	Used to build DimStaff and also for store manager names.
address, city, country	Provide geographic information for DimCustomer and DimStore.

5.2 Transform

During transformation, the extracted data is cleaned and reshaped to fit the star schema. The main transformations are:

- Joining related tables: address, city, and country are joined together to add geographic information to DimCustomer and DimStore.
- Combining names: customer and staff first_name and last_name are joined into a single Full_Name field.
- Generating surrogate keys: each dimension row gets a new sequential key (Customer_Key, Film_Key, etc.). The original OLTP IDs are also kept for reference.
- Building the date dimension: distinct rental, return, and payment dates are collected, and Day, Month, Quarter, Year, Day_Of_Week, and Is_Weekend are derived. The Date_Key is created as YYYYMMDD.
- Handling many-to-many: film_category is many-to-many in theory, but the ETL deduplicates films by Film_ID so that DimFilm has one row per film.
- Standardizing text: city and country names are cleaned and converted to title case.
- Building lookup maps: after dimensions are loaded, lookup dictionaries translate OLTP IDs into the new surrogate keys when building fact tables.

5.3 Load

Loading is done in a specific order. Dimensions are loaded first because fact tables need their surrogate keys. The order is:

- 1. DimDate
- 2. DimCustomer
- 3. DimFilm
- 4. DimStore
- 5. DimStaff
- 6. FactRental
- 7. FactPayment
- 8. FactInventory

New rows are inserted into the warehouse. Fact tables reference the surrogate keys generated by the dimension load step. Foreign key constraints are checked by the database to make sure all references are valid. Data quality checks are run during transformation and loading to make sure the data is clean and consistent. **Slowly Changing Dimensions.** The current design uses SCD Type 1 — when a customer's address or a film's rental rate changes in the OLTP source, the dimension row is overwritten with the new value. For tracking historical changes, the design could be extended to SCD Type 2 by adding `Effective_Date`, `End_Date`, and `Is_Current` columns to the relevant dimensions. This would allow analyses such as comparing a customer's rental behavior before and after they moved to a new city.

6. Data Quality Considerations

Data quality checks are important to make sure that reports are accurate and reliable. The following data quality rules are applied in the ETL process:

- Customer_ID and other business keys must not be null.
- Payment amounts must be greater than zero.
- Duplicate rows in dimension tables (for example duplicate films) must be removed.
- Foreign keys in fact tables must match valid surrogate keys in the dimension tables. Rows that do not match are dropped with a warning.
- Text values such as city and country must be cleaned and standardized (trim whitespace, title case).
- Dates must be in a consistent format across the warehouse.

7. Conclusion

This assignment proposed a high-level data warehouse for a movie rental business based on the Sakila OLTP database. The warehouse uses a star schema with three fact tables (FactRental, FactPayment, FactInventory) and five shared dimensions (DimCustomer, DimFilm, DimStore, DimStaff, DimDate).

An ETL process was designed to extract data from the OLTP source, transform it into the dimensional model, and load it into the warehouse in the correct order. Data quality checks were included to make sure the data is consistent and reliable.

The proposed warehouse supports the analytical business questions defined at the start of the project. It allows managers to study rental activity, revenue, customer behavior, film popularity, store and staff performance, and trends over time. By reorganizing the operational data into facts and dimensions, the warehouse makes reporting easier, faster, and more consistent than running the same queries on the OLTP system.